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Prevention of fire by absorbing local latent heat of fixing belt

Abstract

A device to prevent fire of a fusing belt having a heater positioned inside includes at least one temperature control member adjacent to one side of the fusing belt and, when the at least one temperature control member detects abnormal heat from the fusing belt based on a temperature of the fusing belt, the at least one temperature control member to cut off a supply of electricity to the heater to prevent the fire. The device also includes a heat absorption member adjacent to the other side of the fusing belt and to absorb latent heat from the fusing belt to lower the temperature of the fusing belt to prevent the fire.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a Continuation Patent Application that claims priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/790,210, filed Jun. 30, 2022, which is a U.S. National Stage Entry under 35 U.S.C. § 371 of International Application No. PCT/US2021/012190, filed Jan. 5, 2021, which claims the priority benefit of Korean Patent Application No. 10-2020-0004040, filed Jan. 13, 2020 in the Korean Intellectual Property Office, the contents of all such applications being hereby incorporated by reference in their entirety and for all purposes as if completely and fully set forth herein.

BACKGROUND

(1) In general, an image forming apparatus such as a copier, a printer apparatus, or a facsimile apparatus using an electrophotographic method includes a fuser for fusing an unfused toner transferred to a printing medium to the printing medium by heating and pressing.

(2) As the printing medium passes through a fixing nip (i.e., fusing nip) formed between a fusing belt (or fixing belt) of the fuser and a pressing roller pressed when brought into contact with the fusing belt, an unfused image is heated, pressed, and stably fused on the printing medium.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram illustrating an image forming apparatus equipped with a device for preventing fire of a fusing belt according to an example;

(2) FIG. 2 is a front view illustrating a device for preventing fire of a fusing belt according to an example;

(3) FIG. 3 is a plan view illustrating a device for preventing fire of a fusing belt according to an example;

(4) FIG. 4 is a left side view illustrating a device for preventing fire of a fusing belt according to an example;

(5) FIG. 5 is a left side view illustrating an example of a device for preventing fire of a fusing belt in a state where a temperature control device is omitted in FIG. 4; and

(6) FIG. 6 is a side view illustrating another example of a heat absorption member of a device for preventing fire of a fusing belt according to an example.

DETAILED DESCRIPTION

(7) Hereinafter, various examples will be described with reference to the drawings. The examples described hereinafter may, however, be embodied in many different forms. In the following description, well-known functions or constructions are not described in detail for clarity of description when it is determined that they are well known to those of ordinary skilled in the art.

(8) It will be understood that when an element is referred to as being “connected to” or “coupled to” another element, the element may be “directly connected or coupled” to the other element, or “indirectly connected or coupled” to the other element, such as being “electrically connected to” the other element, with intervening elements therebetween. The terms “comprises,” “comprising,” “includes,” and/or “including” when used herein, specify the presence of components, but do not preclude the presence or addition of one or more other components, unless otherwise specified.

(9) In this disclosure, an “image forming job” may refer to various jobs related to an image (e.g., printing, scanning, or faxing) including forming of an image or creation/storage/transmission of an image file. A “job” may refer to not only an image forming job, but also include a series of processes of performing an image forming job.

(10) In addition, an “image forming apparatus” refers to an apparatus for printing print data generated by a terminal device such as a computer on a printing medium. Examples of such an image forming apparatus include a copier, a printer, a facsimile, or a multi-function printer (MFP) that implements functions of aforementioned devices in a single device.

(11) FIG. 1 is a schematic view illustrating an image forming apparatus equipped with a device for preventing fire of a fusing belt according to an example.

(12) In FIG. 1, the thick solid line indicated by reference numeral P represents a traveling path of a printing medium. In FIG. 1, a main body (no reference numeral) of the image forming apparatus 1 may be a portion forming an external appearance of the image forming apparatus 1.

(13) A paper feeding device 10 may store the printing medium such as paper. The printing medium is conveyed along the traveling path P by a plurality of conveying rollers 11. A charging device 20

may charge a photosensitive body **30** with a predetermined potential. An optical scanning device **40** may scan light to the photosensitive body **30** to form an electrostatic latent image corresponding to print data on the photosensitive body **30**. The charging device **20**, the photosensitive body **30**, and the optical scanning device **40** may configure a print engine for forming an image on the printing medium.

(14) An example of a print engine is now described. A developing device **50** may supply toner to the photosensitive body **30** with the electrostatic latent image formed thereon to form a toner image. The developing device **50** may include a toner accommodating part **51**, a toner supply roller **52**, and a developing roller **53**.

(15) The toner accommodating part **51** accommodates the toner therein. The toner supply roller **52** supplies the toner accommodated in the toner accommodating part **51** to the developing roller **53**, and accordingly, a toner layer is formed on the developing roller **53**. A regulating blade (not shown) makes such a toner layer uniform. The toner layer on the developing roller **53** moves to the electrostatic latent image formed on the photosensitive body **30** due to a potential difference to develop the toner image.

(16) A transfer device **60** may transfer the toner image formed on the photosensitive body **30** to the printing medium. A cleaning device **70** may remove the toner remaining in the photosensitive body **30** after the transfer process is performed.

(17) A fuser **90** may fuse to the printing medium the toner image transferred to the printing medium. The printing medium on which the toner image is fused is discharged out of the image forming apparatus **1** by the plurality of conveying rollers **11**, thereby completing the printing process.

(18) The fuser **90** may include a pressing roller **91** and a fusing belt **93**. In a section where the pressing roller **91** and the fusing belt **93** are in contact with each other, a fixing nip N is formed to extend in a length direction of traveling path P. The fixing nip N is formed to be equal to or larger than a width of the printing medium. An unfused toner, which forms a toner image, is present on the printing medium passing through the transfer device **60**. As heat and pressure are applied to the printing medium while the printing medium passes through the fixing nip N, the unfused toner may be fused.

(19) The pressing roller **91** may be formed of an elastic material such as, for example, rubber or sponge. The pressing roller **91** may apply pressure to the printing medium passing through the fixing nip N. For example, the pressing roller **91** may be pressed toward the fusing belt **93** by an elastic member (not shown).

(20) A heater **95** is disposed inside the fusing belt **93**. The heater **95** may heat a nip forming frame **94** to heat the printing medium passing through the fixing nip N. The nip forming frame **94** may be disposed to be in contact with an inner side of the fusing belt **93**. The heater **95** may be disposed symmetrically with respect to a center of the fusing belt **93** (see FIG. 2).

(21) The fusing belt **93** may be formed of a material having flexibility and may have a width equal to or larger than that of the printing medium. The fusing belt **93** may span the nip forming frame **94** with no tension to form a closed loop.

(22) The fusing belt **93** may be rotated by a frictional force generated between the pressing roller **91** and the fusing belt **93** as the pressing roller **91** rotates. Accordingly, the printing medium passing through the transfer device **60** may pass through the fixing nip N.

(23) The fusing belt **93** is a cylindrical endless belt and may be formed, for example, mainly of a resin film or a metal sleeve. Although not shown in the drawing, the fusing belt **93** may include a base layer and a release layer sheathing one surface of the base layer adjacent to the pressing roller **91** or opposing surfaces of the base layer. To improve image quality of printed matter, an elastic layer may be disposed between the base layer and the release layer to form a relatively wide and flat fixing nip N.

(24) The base layer of the fusing belt **93** is formed of a heat resistant resin such as polyimide,

polyamide, polyimideamide, or a metal such as SUS (Steel Use Stainless), nickel, copper, or the like and may have a thickness of 30 to 200 μm . An appropriate thickness of the base layer may be 50 to 100 μm .

(25) The release layer of the fusing belt **93** may be formed, for example, of a fluorine-based resin such as perfluoroalkoxy (PFA), polytetrafluoroethylene (PTFE), a copolymer of tetrafluoroethylene and hexafluoroethylene (fluorinated ethylene propylene (FEP)), and an appropriate thickness thereof is about 10 to 30 μm . As a material for the release layer, fluorine-based resins are mainly used, and the thickness is, for example, 10 to 50 μm . As the fluorine-based resin, perfluoroalkyl, polytetrafluoroethylene, a copolymer of tetrafluoroethylene and hexafluoroethylene, and the like, for example, may be used. As the release layer, a tube formed of a fluorine resin may be used, for example, and the release layer may be formed by a coating method using the fluorine resin.

(26) As the elastic layer of the fusing belt **93**, fluorine rubber, silicone rubber, or the like, for example, may be used. In the elastic layer, an insulating elastic layer may include various rubber materials such as, for example, fluorine rubber, silicone rubber, natural rubber, isoprene rubber, butadiene rubber, nitrile rubber, chloroprene rubber, butyl rubber, acrylic rubber, hydrin rubber, urethane rubber, and various elastic material such as, for example, thermoplastic elastomer such as styrenes, polyolefins, polyvinyl chlorides, polyurethanes, polyesters, polyamides, polybutadienes, toransporisoprenes, chlorinated polyethylenes, and the like, and may include one or two or more kinds thereof.

(27) The fuser **90** provided in the image forming apparatus of the disclosure uses heat to melt the toner and fuses toner to paper, in which case the heat is generated from the heater **95**. By making the fusing belt **93** thinner and increasing the efficiency of the heater **95**, the printing speed is increased and the print time of a first sheet is shortened.

(28) As the efficiency of the heater **95** increases with the reduction in the thickness of the fusing belt **93** described above, a time for the fusing belt **93** to catch fire due to abnormal overheating is gradually shortened and the fusing belt **93** would catch fire more quickly. To prevent such fire, the image forming apparatus **1** of the disclosure includes a device **100** for preventing fire of the fusing belt at a position adjacent to the fusing belt **93**. The device **100** for preventing fire of the fusing belt is to detect whether the fusing belt has reached a temperature at which the fusing belt may ignite a fire to cut off power supplied to the heater **95** to turn off the heater **95**, and absorb latent heat generated locally at the fusing belt **93**, thereby fundamentally preventing firing due to the latent heat.

(29) Hereinafter, an example of a configuration of the device **100** for preventing fire of a fusing belt according to the disclosure will be described in detail with reference to the drawings.

(30) FIGS. **2** to **4** are a front view, a plan view, and a left side view, respectively, showing a device **100** for preventing fire of a fusing belt according to an example. FIG. **5** is a left side view showing a device for preventing fire of a fusing belt in an example in which temperature control members **110** and **120** shown in FIG. **4** are omitted.

(31) Referring to FIGS. **2** to **4**, the device **100** for preventing fire of the fusing belt includes first and second temperature control members **110** and **120** disposed adjacent to one side of the fusing belt **93**, a heat absorption member **130** disposed adjacent to the other side of the fusing belt **93**, and first and second temperature sensors **161** and **162** for detecting a temperature of the fusing belt **93**.

(32) In this disclosure, the first and second temperature control members **110** and **120** and the first and second temperature sensors **161** and **162** are provided as pairs, respectively, but the disclosure is not limited thereto and at least one of each of the temperature control members and temperature sensors may be provided. In addition, in this disclosure, one heat absorption member **130** is provided, but the disclosure is not limited thereto and the heat absorption member **130** may be manufactured in a smaller size and two or more heat absorption members **130** may be provided.

(33) The first and second temperature control members **110** and **120** may be thermostats. When a temperature of the fusing belt **93** rises above a predetermined temperature due to, for example,

abnormal heat generated in the heater **95**, the first and second temperature control members **110** and **120** may cut off supply of electricity to the heater **95** to prevent the heater **95** from generating more heat. The first and second temperature control members **110** and **120** may include a bimetal, for example, and therefore permanently cut off supply of electricity to the heater **95** when a temperature of the bimetal is higher than or equal to a threshold value.

(34) The fusing belt **93** may locally swell due to occurrence of abnormal heat of the heater **95**. In this case, the fusing belt **93** may ignite, or otherwise react, due to the abnormal heat in a time shorter than a time in which the fusing belt **93** may ignite, or otherwise react, due to radiant heat and convective heat. The first and second temperature control members **110** and **120** may be spaced apart from each other so as not to come into contact with the fusing belt **93** due to movement of the fusing belt **93** that may take place during a normal operation of the fusing belt **93**. In this case, the first and second temperature control members **110** and **120** may be arranged to maintain first and second gaps **G1** and **G2** from an outer surface of the fusing belt **93**, respectively.

(35) According to the arrangement of the first and second temperature control members **110** and **120** with respect to the fusing belt **93**, when one side of the fusing belt **93** is swollen by local heat and brought into contact with at least one of the first and second temperature control members **110** and **120**, for example in the case of abnormal heat generation of the heater **95**, the bimetals of the first and second temperature control members **110** and **120** may open due to conductive heat and react more quickly to quickly cut off supply of electricity to the heater **95** than the first and second temperature control members **110** and **120** would react to radiant heat and convective heat.

(36) As such, a portion of the entire area of the fusing belt **93** swollen by abnormal heat generation of the heater **95** may vary according to a position of the heater **95** inside the fusing belt **93**. That is, referring to FIG. **4**, the nip forming frame **94** is disposed at a position adjacent to the pressing roller **91** inside the fusing belt **93**. In this case, the heater **95** is disposed at a position away from the pressing roller **91** so as not to interfere with the nip forming frame **94**. Therefore, in this disclosure, the first and second temperature control members **110** and **120** may be disposed adjacent to an upper side of the fusing belt **93** closest to the heater **95** positioned inside the fusing belt **93**. As such, the first and second temperature control members **110** and **120** are disposed, for example, in consideration of the position of the heater **95**.

(37) The first temperature control member **110** has a positive (+) terminal **111** and a negative (-) terminal **113** to which a wire (not shown) is connected, and the second temperature control member **120** also has a positive (+) terminal **121** and a negative (-) terminal **123**. Each of the terminals **111**, **113**, **121**, and **123** of the first and second temperature control members **110** and **120** may be connected to wires and supported by a structure (not shown) disposed near the fusing belt. Due to the structure, the first and second temperature control members **110** and **120** may be arranged to maintain desired gaps, e.g., first and second gaps **G1** and **G2**, from the outer surface of the fusing belt **93**.

(38) Meanwhile, even when the heater **95** is turned off as the supply of electricity to the heater **95** is cut off by the first and second temperature control members **110** and **120**, local latent heat may occur on the other side of the fusing belt **93**, i.e., on the other side of the fusing belt where the first and second temperature sensors **161** and **162** are arranged. The fusing belt **93** may burn out due to local latent heat. The heat absorption member **130** may absorb the local latent heat of the fusing belt **93** and disperse the absorbed local latent heat around the fusing belt **93** to lower a temperature of the fusing belt **93**, thereby eliminating a fire factor.

(39) The heat absorption member **130** may be formed of, for example, a metal material (e.g., aluminum, copper, etc.) having excellent thermal conductivity. To reduce a manufacturing cost of the heat absorption member **130**, the heat absorption member **130** may be manufactured with relatively inexpensive aluminum.

(40) Referring to FIGS. **3** and **5**, the heat absorption member **130** may be disposed on the other side of the fusing belt **93** and maintain a predetermined gap from the outer surface of the fusing belt **93**.

The heat absorption member **130** may be formed by bending a flat plate in multiple stages to surround a portion of the fusing belt **93** along a width direction (which refers to a direction perpendicular to a length direction of the fusing belt **93**) of the fusing belt **93**.

(41) A region where the heat absorption member **130** is disposed may vary depending on a position of the heater **95** inside the fusing belt **93**, similarly to the positions of the first and second temperature control members **110** and **120** described above. That is, referring to FIG. 5, the heater **95** is disposed at a position away from the pressing roller **91** to avoid interfering with the nip forming frame **94**. Therefore, in this disclosure, the heat absorption member **130** may be disposed above the fusing belt **93** closest to the heater **95** positioned inside the fusing belt **93**. As such, the heat absorption member **130** may be disposed in consideration of the position of the heater **95**.

(42) The heat absorption member **130** may include a first heat absorption part **131** and a second heat absorption part **132** which are brought into contact with a swollen portion of the fusing belt **93** due to local latent heat. The first heat absorption part **131** may be disposed with a third gap **G3** with respect to the outer surface of the fusing belt **93**, and the second heat absorption part **132** may be disposed with a fourth gap **G4** with respect to the outer surface of the fusing belt **93**.

(43) As the second heat absorption part **132** extends to one side of the first heat absorption part **131** and is bent at a predetermined angle with respect to the first heat absorption part **131**, the second heat absorption part **132** may be positioned at a downstream portion with respect to the first heat absorption part **131** with reference to the width direction of the fusing belt **93**. The first and second heat absorption parts **131** and **132** may come into contact with the fusing belt **93** that swells at different points along the width direction of the fusing belt **93**.

(44) The third and fourth gaps **G3** and **G4** are distances at which the first and second heat absorption parts **131** and **132** are spaced apart from the fusing belt **93** and not to be in contact with the fusing belt **93** due to movement of the fusing belt **93** that takes place during a normal operation of the fusing belt **93**, which also counts a distance at which a swollen portion of the fusing belt **93** due to abnormal heat generation of the heater **95** comes into contact with the first and second heat absorption parts **131** and **132**. The first and second heat absorption parts **131** and **132** may absorb local latent heat of the fusing belt **93** through conduction and lower a temperature of the fusing belt **93**, thereby fundamentally preventing fire of the fusing belt **93** due to latent heat.

(45) In this disclosure, the heat absorption part is described as two bent parts, but the disclosure is not limited thereto and the heat absorption part may have one or at least three parts. If the heat absorption part has at least three parts, the heat absorption part may be bent in multiple stages at regular intervals along a profile of the fusing belt **93**.

(46) The heat absorption member **130** includes a fixed part **134** extending indirectly from the second heat absorption part **132** through a connection part **133**. The fixed part **134** is fixed to a peripheral structure **200** through a fastening screw (not shown) so that the heat absorption member **130** may be disposed adjacent to the fusing belt **93**. The fixed part **134** may have a through hole **134a** (see FIG. 3) through which the fastening screw passes.

(47) The heat absorption member **130** may further include first to third additional heat absorption parts **135**, **136**, and **137** to increase a heat absorption area. The first to third additional heat absorption parts **135**, **136**, and **137** serve to increase the heat absorption area and absorb heat from the fusing belt **93** when a swollen portion of the fusing belt **93** due to abnormal heat generation of the heater **95** comes into contact therewith. Accordingly, the heat absorption member **130** may extend a heat absorption point with respect to the fusing belt **93** through the first to third additional heat absorption parts **135**, **136**, and **137**.

(48) The first to third additional heat absorption parts **135**, **136**, and **137** may be arranged not to interfere with a structure or component installed around the heat absorption member **130**, e.g., the first and second temperature sensors **161** and **162**. To this end, the first to third additional heat absorption parts **135**, **136**, and **137** may extend with a predetermined length toward one side of the fusing belt **93** along the length direction of the fusing belt **93** from the other side of the first heat

absorption part **131** as shown in FIG. 3.

(49) The first to third additional heat absorption parts **135**, **136**, and **137** may be arranged at right angles with respect to the first and second heat absorption parts **131** and **132**. However, a direction in which the first to third additional heat absorption parts **135**, **136**, and **137** extend may face in various directions in consideration of a peripheral structure of the heat absorption member **130** or a component to be installed.

(50) The first additional heat absorption part **135** may be coplanar with the first heat absorption part **131**. The second and third additional heat absorption parts **136** and **137** may be bent in multiple stages along the width direction of the fusing belt **93** from the first additional heat absorption part **135**. The second and third additional heat absorption parts **136** and **137** may be bent at an appropriate angle so as not to interfere with a peripheral structure of the heat absorption member **130** or the like.

(51) The third additional heat absorption part **137** may have fixed pieces **137a**, which may be inserted and fixed to a peripheral structure or the like, extending from both sides thereof to firmly fix the heat absorption member **130**. In this case, the fixed pieces **137a** need not be a pair and at least one fixed piece **137a** may be provided. In addition, a shape of the fixed piece **137a** is not limited to a specific shape and may be formed in consideration of various shapes of a peripheral structure.

(52) Meanwhile, in this disclosure, the heat absorption member **130** further includes the first to third additional heat absorption parts **135**, **136**, and **137**, but the number of the additional heat absorption parts is not limited thereto and the heat absorption member **130** may have at least four multi-bent shape. In this manner, as the number of additional heat absorption parts increases, the heat absorption area of the heat absorption member **130** may be increased.

(53) In addition, the heat absorption member **130** has a simple structure to be easily manufactured, and may be easily installed not to interfere with a peripheral structure when installed near the fusing belt **93**. The heat absorption member **130** may be manufactured with aluminum which is low in price among metals having high thermal conductivity to lower a manufacturing cost.

(54) The first and second temperature sensors **161** and **162** may be thermistors and may be disposed on the other side of the fusing belt **93**. Each of the temperature sensors **161** and **162** may be disposed to maintain a predetermined gap with the fusing belt **93** not to interfere with rotational driving of the fusing belt **93**.

(55) In this disclosure, heat of a portion of the fusing belt **93** where preceding fire occurs is absorbed through the heat absorption member **130** configured as described above at a portion where the temperature is locally highest due to a heating pattern of the heater **95**, thereby preventing fire in advance, as well as delaying a fire start time.

(56) The first temperature sensor **161** may measure a temperature of the fusing belt **93** at a position different from the second temperature sensor **162** in the length direction of the fusing belt **93**. For example, the first temperature sensor **161** may be installed to be positioned closer to a central portion of the fusing belt **93** than the second temperature sensor **162** in the length direction of the fusing belt **93**.

(57) In addition, when the first temperature sensor **161** is disposed at a first position in the width direction of the fusing belt **93**, the second temperature sensor **162** may be disposed at a second position which is a downstream side with respect to the first temperature sensor **161** to measure temperatures at different points of the fusing belt **93**.

(58) As described above, in this disclosure, whether the fusing belt **93** is overheated may be reliably determined by utilizing temperatures measured at different positions with respect to the fusing belt **93** through the respective temperature sensors **161** and **162**.

(59) The first and second temperature sensors **161** and **162** may indirectly measure a temperature of the heater **95** by measuring a temperature of the fusing belt **93**. The first and second temperature sensors **161** and **162** may transmit the measured temperatures to a controller (not shown) and the

controller may control a driving state of the image forming apparatus **1**. For example, when the temperatures measured by the first and second temperature sensors **161** and **162** are higher than a reference value of the temperature of the heater **95**, the controller turns off the heater **95** to control the image forming apparatus **1** to cool the fusing belt **93** while being idly rotated.

(60) FIG. **6** is a side view illustrating another example of a heat absorption member of a device of preventing fire of the fusing belt according to an example.

(61) The heat absorption member **130** described above is formed to be bent in multiple stages to surround the portion of the fusing belt **93**. However, a heat absorption member **230** is not limited to such a shape and may be provided such that a portion thereof surrounding the fusing belt **93** corresponds to a profile of the fusing belt **93**.

(62) A heat absorption part **231** of the heat absorption member **230** may be provided as a curved surface to correspond to a profile of the fusing belt **93** configured as a curved surface. The heat absorption part **231** may be disposed with a fifth gap **G5** and a sixth gap **G6** with respect to two points of an outer surface of the fusing belt **93**, similar to the first and second heat absorption parts **131** and **132** described above. Accordingly, the heat absorption part **231** may come into contact with the fusing belt **93** swollen at different points along the width direction of the fusing belt **93**.

(63) The fifth and sixth gaps **G5** and **G6** are distances at which the heat absorption part **231** is spaced apart from the fusing belt **93** and not to be in contact with the fusing belt **93** due to movement of the fusing belt **93** that takes place during a normal operation of the fusing belt **93**, which also counts a distance at which a swollen portion of the fusing belt **93** due to abnormal heat generation of the heater **95** comes into contact with the heat absorption part **231**. The heat absorption part **231** may absorb local latent heat of the fusing belt **93** through conduction and lower a temperature of the fusing belt **93**, thereby fundamentally preventing fire of the fusing belt **93** due to latent heat.

(64) The heat absorption member **230** may include an extending part **233** and a fixed part **234** extending from one side of the heat absorption part **231**. The fixed part **234** may be fixed to a peripheral structure through a fastening screw. The extending part **233** and the fixed part **234** may be bent in multiple stages and a connection part connecting the extending part **233** and the fixed part **234** may be formed in a curve having a predetermined curvature.

(65) The heat absorption member **230** may further include first to third additional heat absorption parts **235** (not shown), **236**, and **237** to increase a heat absorption area. The first to third additional heat absorption parts **235**, **236**, and **237** may have the same configuration as that of the first to third additional heat absorption parts **135**, **136**, and **137** of the heat absorption member **130**.

(66) That is, the first additional heat absorption part **235** may be coplanar with the heat absorption part **231**. The second and third additional heat absorption parts **236** and **237** may be bent in multiple stages along the width direction of the fusing belt **93** from the first additional heat absorbing part **235**. The second and third additional heat absorption parts **236** and **237** may be bent at an appropriate angle so as not to interfere with a peripheral structure of the heat absorption member **230**. The third additional heat absorption part **237** may have fixed pieces **237a**, which may be inserted and fixed to a peripheral structure or the like, extending from both sides thereof to firmly fix the heat absorption member **230**.

(67) As above, the present invention has been illustrated and described with respect to preferred examples of the present invention, but the present invention is not limited to the above-described examples and various modifications may be made by a person skilled in the art to which the present invention pertains, without departing from the spirit of the invention as defined by the claims of the present invention, and such modifications also fall within the claims.

Claims

1. A device comprising: a thermostat adjacent a first side of a fusing belt, wherein the thermostat is to detect heat from the fusing belt and to cut off electricity to a heater when the detected heat exceeds a threshold; and a thermal conductor to absorb heat from the fusing belt to lower the temperature of the fusing belt, wherein the thermal conductor is located entirely on an opposite side of the thermostat with respect to a center of the fusing belt in an axial direction of the fusing belt, wherein the thermostat and the thermal conductor are overlapping when viewed in the axial direction.
2. The device of claim 1, wherein the thermal conductor maintains a predetermined gap from an outer surface of the fusing belt.
3. The device of claim 1, wherein the thermal conductor includes a curved surface portion corresponding to a profile of the fusing belt.
4. The device of claim 1, wherein the thermal conductor is bent in multiple stages.
5. The device of claim 1, wherein the thermal conductor comprises: a fixed part fixed to a peripheral structure; and a heat absorption part extending from the fixed part and having a predetermined length along a width direction of the fusing belt.
6. The device of claim 5 wherein an area of the heat absorption part is maintained at a first interval from the fusing belt.
7. The device of claim 6, wherein the thermal conductor further comprises an additional heat absorption part extending from the heat absorption part.
8. The device of claim 7, wherein an area of the additional heat absorption part is maintained at the first interval from the fusing belt.
9. The device of claim 7, wherein the additional heat absorption part has a predetermined length along a length direction of the fusing belt.
10. The device of claim 7, wherein the heat absorption part and the additional heat absorption part are at a right angle.
11. The device of claim 1, wherein the thermal conductor comprises aluminum.
12. An image forming apparatus comprising: a main body; a print engine in the main body and to form an image on a printing medium; a fuser including a pressing roller and a fusing belt to face the pressing roller and to allow a heater to be included inside thereof; and a device to prevent fire of the fusing belt, the device including: a thermostat adjacent a first side of a fusing belt, wherein the thermostat is to detect heat from the fusing belt and to cut off electricity to a heater when the detected heat exceeds a threshold; and a thermal conductor to absorb heat from the fusing belt to lower the temperature of the fusing belt, wherein the thermal conductor is located entirely on an opposite side of the thermostat with respect to a center of the fusing belt in an axial direction of the fusing belt, wherein the thermostat and the thermal conductor are overlapping when viewed in the axial direction.
13. The image forming apparatus of claim 12, wherein each of the thermostat and the thermal conductor is at the same interval with respect to the fusing belt.
14. The image forming apparatus of claim 12, wherein the thermal conductor includes a curved surface portion corresponding to a profile of the fusing belt.
15. The image forming apparatus of claim 12, wherein the thermal conductor is bent in multiple stages.
16. The image forming apparatus of claim 12, wherein the thermal conductor comprises: a fixed part fixed to a peripheral structure; and a heat absorption part extending from the fixed part and having a predetermined length along a width direction of the fusing belt.
17. The image forming apparatus of claim 16, wherein an area of the heat absorption part is maintained at a first interval from the fusing belt.
18. The image forming apparatus of claim 17, wherein an area of the additional heat absorption part is maintained at the first interval from the fusing belt.

19. The image forming apparatus of claim 17, wherein the heat absorption part and the additional heat absorption part are at a right angle.
20. The image forming apparatus of claim 16, wherein the thermal conductor further comprises an additional heat absorption part extending from the heat absorption part.
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